ADITYA UNIVERSITY

SCHOOL OF ENGINEERING

Department of Data Science

CERTIFICATE

This is to certify that thve Cornerstone project work entitled *"**E-Commerce Simulator**"* is submitted by

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in partial fulfillment of the requirements for the award of the B.Tech. degree in Data Science for the academic year 2025-2026.

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DECLARATION

We hereby declare that the Cornerstone project entitled "**E- Commerce Simulator**" is original and is submitted to **ADITYA UNIVERSITY**, Surampalem, in partial fulfillment of the requirements for the award of the **B.Tech. degree** in **Data Science**. We further declare that this project work has not been submitted, either in full or in part, for the award of any degree in this or any other institution. We also confirm that this work complies with the Institutional Academic Integrity and AI Usage Policy

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ABSTRACT

The E-Commerce Application is a desktop-based system designed to efficiently manage product information, cart operations, and order processing by replacing manual or unstructured methods with a digital solution. The system enables users to perform operations such as viewing products, adding items to the cart, updating quantities, placing orders, and applying coupon-based discounts in a structured and reliable manner using a relational database.

Developed using Core Java, Swing for the graphical user interface, JDBC for database connectivity, and MySQL for backend storage, the application ensures data accuracy, reduces redundancy, and improves accessibility. The system also supports product management functionalities, allowing sellers to add, update, and delete product records efficiently.

With its user-friendly interface and efficient data handling capabilities, the application enhances the overall shopping experience and demonstrates the core functionalities of an online e-commerce system. It provides a reliable and scalable solution that can be further extended with advanced features in the future.

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ACRONYMS

- **JDBC** : Java Database Connectivity
- **GUI** : Graphical User Interface
- **SQL** : Structured Query Language

1. INTRODUCTION

1.1 Brief Information about the Project

The Online Shopping Cart Simulation is a Java-based application developed to simulate an online shopping platform with two types of users: Buyers and Sellers. The system allows buyers to browse products and place orders, while sellers can manage product listings and inventory. The system demonstrates real-world e-commerce concepts such as role-based access, inventory management, and order processing.

This application is developed using:

1 Java (Core Java)

2 Swing (GUI)

3 JDBC

4 Oracle Database

1.2 Motivation and Contribution of Project

In many small-scale systems, product and order data are often managed manually or using basic tools, which leads to inefficiencies such as data inconsistency, difficulty in tracking orders, and increased chances of errors or data loss. These limitations motivated the development of the E-Commerce Application.

The key contributions of this project include:

• Providing a digital solution for managing products and orders

• Reducing manual effort in handling cart and transaction data

• Improving accuracy and consistency of stored information

• Enabling quick access to product and order details

• Enhancing user experience through a graphical interface

• Implementing coupon-based discount functionality

This system simplifies the process of online shopping and improves overall efficiency in managing product and order data.

1.3 Objectives of the Project

The main objectives of the E-Commerce Application are:

• To develop a user-friendly application for online shopping

• To store product, cart, and order information in a structured database

• To provide functionalities for adding, updating, and deleting products

• To enable users to add items to cart and manage quantities

• To allow users to place orders and view order details

- To implement coupon-based discounts for enhanced functionality
- To integrate the application with a database using JDBC

1.4 Scope of the Project

The E-Commerce Application can be used in small-scale environments such as academic projects, prototype systems, or small businesses to manage products and orders efficiently. It supports operations such as viewing products, adding items to cart, managing cart contents, placing orders, applying coupons, and displaying order details in an organized manner.

The system provides a simple and user-friendly graphical interface for smooth navigation between different functionalities. It maintains product and transaction data in a centralized database, ensuring consistency and easy access. The current system focuses on core e-commerce features and can be further extended with advanced functionalities such as payment integration, user authentication, and recommendation systems.

2. SYSTEM ANALYSIS

2.1 Functional Requirements

Functional requirements define the core operations and functionalities that the E-Commerce System must perform to enable users to browse products, manage cart items, place orders, and apply discounts. The functional requirements of the E-Commerce System are as follows:

• **View Products:** The system should allow users to view available products along with details such as product name, price, category, description, and image. The product information should be displayed in a clear and user-friendly format.

• **Add Products to Cart:** The system should enable users to add selected products to their cart. It should prevent duplicate entries of the same product and instead update the quantity if the product already exists in the cart.The system should allow users to view items in their cart, update product quantities, and remove items if required. The cart should dynamically reflect changes made by the user.

• **Place Orders:** The system should allow users to place orders based on the items present in the cart. It should store order details such as product ID, user ID, price at the time of purchase, and quantity. The system should allow users to apply coupon codes during checkout. It should validate the coupon, check eligibility conditions such as minimum purchase amount, and apply the appropriate discount.

• **View Order Details**: The system should provide functionality to view details of previously placed orders, including product information, price, quantity, and seller details.

• **Product Management (Seller Side):** The system should allow sellers to add new products, update existing product details, and delete products. Proper validation should be applied to ensure data accuracy.

2.2 Non-Functional Requirements

Non-functional requirements define the quality attributes and performance standards that the E-Commerce System must satisfy.

• **Performance**: The system should provide quick response times for operations such as product loading, cart updates, and order placement without noticeable delays.

• **Security**: The system should ensure that only authorized users can perform operations such as modifying products or placing orders. Database operations should be handled securely using prepared statements.

• **Usability:** The system should provide a simple and intuitive graphical user interface using Java Swing, allowing users to easily navigate between product, cart, and order pages.

• **Reliability:** The system should function consistently without crashes and handle errors gracefully, ensuring that no data is lost during operations.

• **Scalability**: The system should be capable of handling an increasing number of products, users, and orders without significant degradation in performance.

• **Maintainability:** The system should be designed in a modular way, allowing easy updates and modifications to features such as cart logic, coupon system, or UI components.

2.3 Requirements Specification

This section specifies the hardware and software needed to run the system.

2.3.1 Minimum Hardware Requirements

- Processor: Intel i3 or above
- RAM: 4 GB (minimum)
- Hard Disk: 20 GB free space
- Monitor: Standard display
- Keyboard and Mouse

2.3.2 Software Requirements

- **Operating System:** Windows 7/10/11
- **Programming Language:** Java
- **IDE:** VS Code
- **Database:** Oracle Database
- **Connectivity:** JDBC

3. TECHNOLOGY DESCRIPTION

1.1 Brief Information about the Project

The E-Commerce Application is a desktop-based system developed to facilitate online product browsing, cart management, and order processing. It provides a centralized platform where users can view products, add items to a cart, place orders, and view order details in an organized manner. The system replaces manual or unstructured methods of managing product and order information with a computerized solution, thereby improving efficiency, reducing errors, and enhancing user experience.

The application is developed using Core Java for programming, Swing for designing the graphical user interface, JDBC for database connectivity, and MySQL for storing product, cart, and order data. The system is designed to be user-friendly and suitable for small-scale e-commerce environments such as academic projects and basic retail systems.

1.2 Motivation and Contribution of Project

In many small-scale systems, product and order management is handled manually or through basic tools, which leads to issues such as data inconsistency, difficulty in tracking orders, and lack of efficient user interaction. These limitations motivated the development of the E-Commerce Application.

The key contributions of this project include:

• Providing a digital platform for managing products and orders

• Reducing manual effort in handling product and cart data

• Improving accuracy and consistency of stored information

• Enabling quick access to product and order details

• Enhancing user experience through a graphical interface

• Introducing coupon-based discount functionality

This system simplifies the process of online shopping and demonstrates how real-world e-commerce applications function.

1.3 Objectives of the Project

The main objectives of the E-Commerce Application are:

• To develop a user-friendly GUI-based shopping system

• To store product, cart, and order data in a structured database

• To implement functionalities for adding, updating, and deleting products

• To enable users to add items to cart and manage quantities

• To allow users to place orders and view order details

• To implement a coupon system for applying discounts

• To integrate Java application with database using JDBC

1.4 Scope of the Project

The E-Commerce Application can be used in small-scale environments such as academic projects, small businesses, or prototype systems to manage products and orders efficiently. It supports operations such as viewing products, adding items to cart, managing cart contents, placing orders, applying coupons, and displaying order details in a structured format.

The system provides a simple and interactive graphical interface, making it easy for users to navigate through different functionalities. While the current implementation focuses on core e-commerce features, it can be extended further with advanced functionalities such as payment integration, user authentication, and recommendation systems.

4. DATABASE DESIGN

4.1 Database Tables

The E- Commerce Simulator mainly has 5 relational database tables in it. Each of them are used to store a particular information. Out of the 5 tables, 4 of them are related to each other and one of them is independent of all.

- Users Table: Stores details of all the users.
- Products table: Stores details of all the products.
- Orders Table: Stores details of all the orders that were placed in the application.
- Cart Table: Stores details of all of the products that were added to the cart by the user.
- Coupon Table: Stores various coupons that can be accessed to get a particular discount.

4.2 Table Structure

4.2.1 Users Table:

Field	Type	Null	Key	Default	Extra
user_id	int	NO	PRI	NULL	auto_increment
user_name	varchar(50)	NO	UNI	NULL	
fullname	varchar(100)	NO		NULL	
password	varchar(100)	NO		NULL	
address	text	YES		NULL	
contact_no	varchar(15)	YES		NULL	
user_type	enum('buyer','seller')	NO		NULL	

Table 1: User Relational Database Table

4.2.2 Product Table:

Field	Type	Null	Key	Default	Extra
product_id	int	NO	PRI	NULL	auto_increment
product_name	varchar(100)	NO		NULL	
description	text	YES		NULL	
price	double	NO		NULL	
category	varchar(50)	YES		NULL	
seller_id	int	NO	MUL	NULL	
pic	varchar(255)	YES		NULL	
stock	int	NO		0	
rating	double	YES		0	

Table 2: Products Relational Database Table

4.2.3 Orders Table:

Field	Type	Null	Key	Default	Extra
order_id	int	NO	PRI	NULL	auto_increment
user_id	int	NO	MUL	NULL	
product_id	int	NO		NULL	
price	double	NO		NULL	
quantity	int	NO		NULL	
order_time	timestamp	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED

Table 3: Orders Relational Database Table

4.2.4 Cart Table:

	Field	Type	Null	Key	Default	Extra
▶	user_id	int	NO	PRI	NULL	
	product_id	int	NO	PRI	NULL	
	quantity	int	NO		NULL	

Table 4: Cart Relational Database Table

4.2.5 Coupons Table:

	Field	Type	Null	Key	Default	Extra
▶	coupon_id	int	NO	PRI	NULL	auto_increment
	code	varchar(20)	NO		NULL	
	validity	datetime	NO		NULL	

Table 5: Coupons Relational Database Table

4.3 Keys and Constraints

Keys and constraints are used to maintain data integrity and ensure accurate data storage.

- **Primary Keys:**
  - **User_id** in users table
  - **Product_id** in products table
  - **Order_id** in orders table
  - **{user_id, product_id}** in cart table
  - **Coupon_id** in coupon table
- **Foreign Key:**
  - **Seller_id** in product table is a foreign key references users_id in user table.
  - **User_id** in Orders table references users_id in user table.
  - **Product_id** in orders table references product_id in products table.
  - **user_id** in carts table references user_id in user table
  - **product_id** in carts table references product_id in products table.
- **NOT NULL Constraint:**
  - Important fields such as user_name, full name, password, user_type in users table and product_name, price and stock are not null by default.
- **Data Type Constraints:**
  - Each field accepts only valid data types to ensure correctness

These constraints help in maintaining consistency and avoiding duplicate or invalid data.

4.4 Table Relationships

There are 4 tables that are related to each other and one independent table.

- Users table stores the details of the user and also related to products table, cart table and orders table.
- Product table stores the products details, and each product has a seller, and seller must be a user. Hence it depends on the users table.
- Cart table stores the details of the carts of every user and fetches details when of the buyer whenever called. Hence its related to both users table and products table.
- Similarly orders table also is related to both products table and users table.

5. SYSTEM DESIGN

5.1 System Architecture

The Employee Management System follows a **three-layer architecture**:

1. **Presentation Layer (User Interface)**
   - Developed using Java Swing
   - Allows users to interact with the system
2. **Application Layer (Business Logic)**
   - Processes user inputs
   - Performs operations like add, update, delete, search
3. **Data Layer (Database)**
   - Stores all the tables data
   - Managed using MySQL Database

This architecture ensures better organization, security, and maintainability.

5.2 Modules Description

The system is divided into the following modules:

1. Product Management Module

- Adds new product details
- Updates existing product information
- Deletes product records
- Displays product details

2. Cart Management Module

- Adds products to cart
- Prevents duplicate product entries
- Updates product quantity
- Displays cart items

3. Order Management Module

- Places orders from cart
- Stores order details (price, quantity)
- Displays order history
- Fetches ordered product details

4. Coupon Module

- Validates coupon codes

- Applies discount based on conditions
- Checks minimum purchase amount
- Calculates final payable amount

5. Database Connectivity Module

- Connects Java application with MySQL database
- Executes SQL queries using JDBC
- Handles insert, update, delete, and select operations

6. User Interface Module

- Provides GUI using Java Swing
- Displays product, cart, and order screens
- Handles user interactions (buttons, forms)

7. Image Handling Module

- Uploads product images
- Displays images in UI
- Resizes images dynamically

8. Navigation Module

- Switches between different screens
- Manages frame transitions
- Handles page flow (product → cart → order)

6. SYSTEM IMPLEMENTATION

Buyer Module:

Login Screen: Displays the login interface where the user enters username, password and user type.

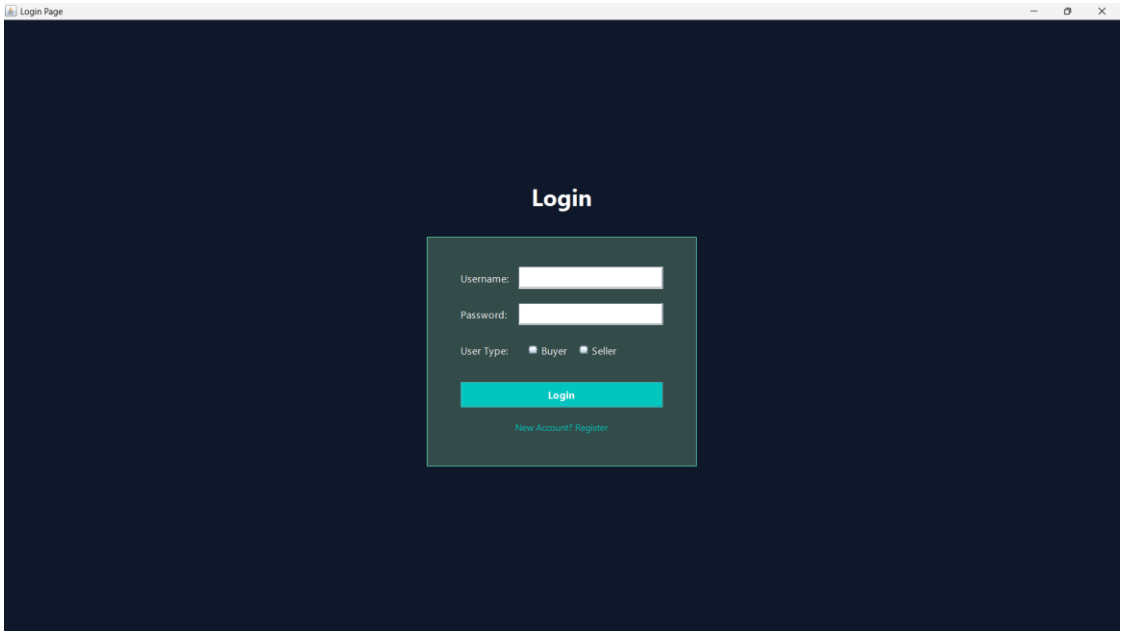

Figure 1: Login Screen of Online E – Commerce simulator

Registration Page: Displays the registration interface where the user registers with his details.

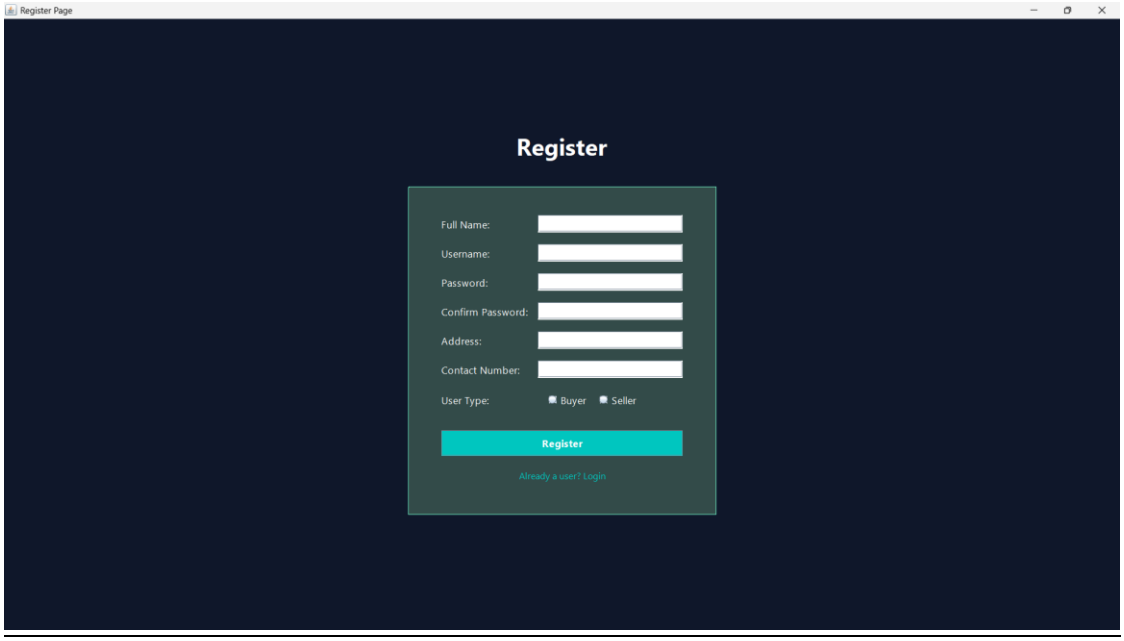

Figure 1: Registration Screen of Online E – Commerce simulator

Welcome Screen: The welcome page shows all the categories and top 5 products in each category.

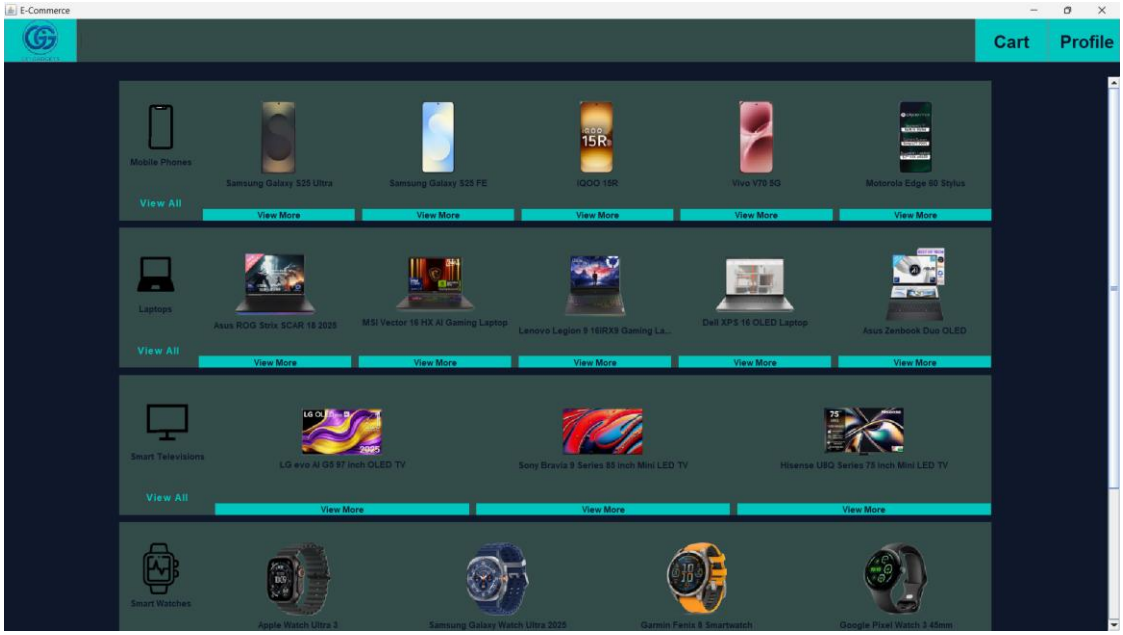

Figure 3: Welcome Page

Product Page: Shows the product details.

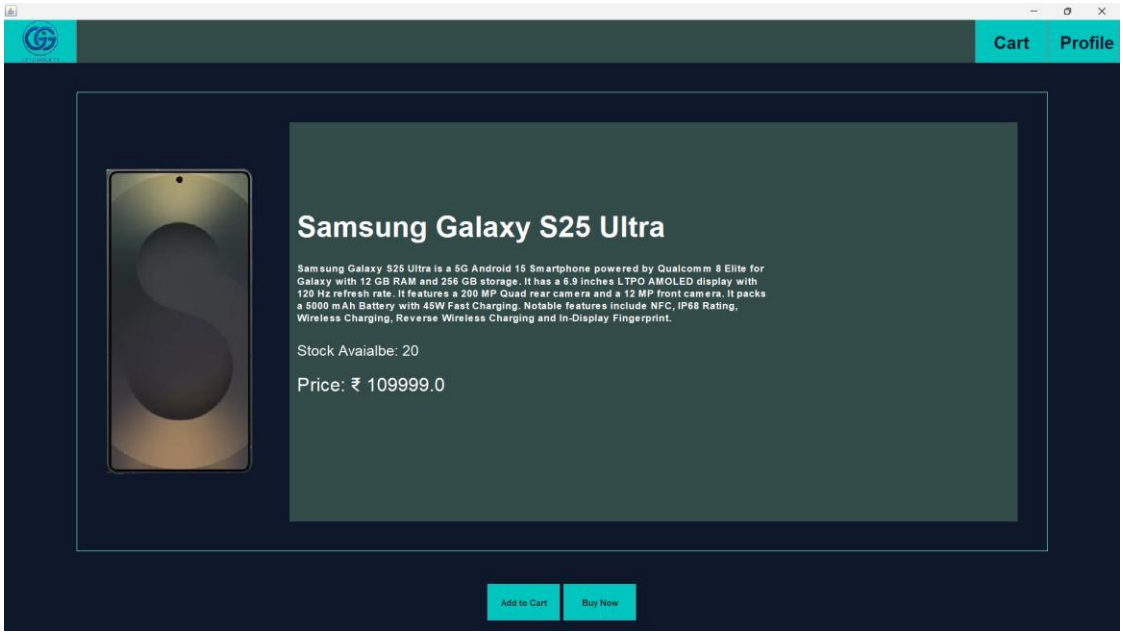

Figure 4: Product Details

Cart Page: All the products added in the cart are shown here.

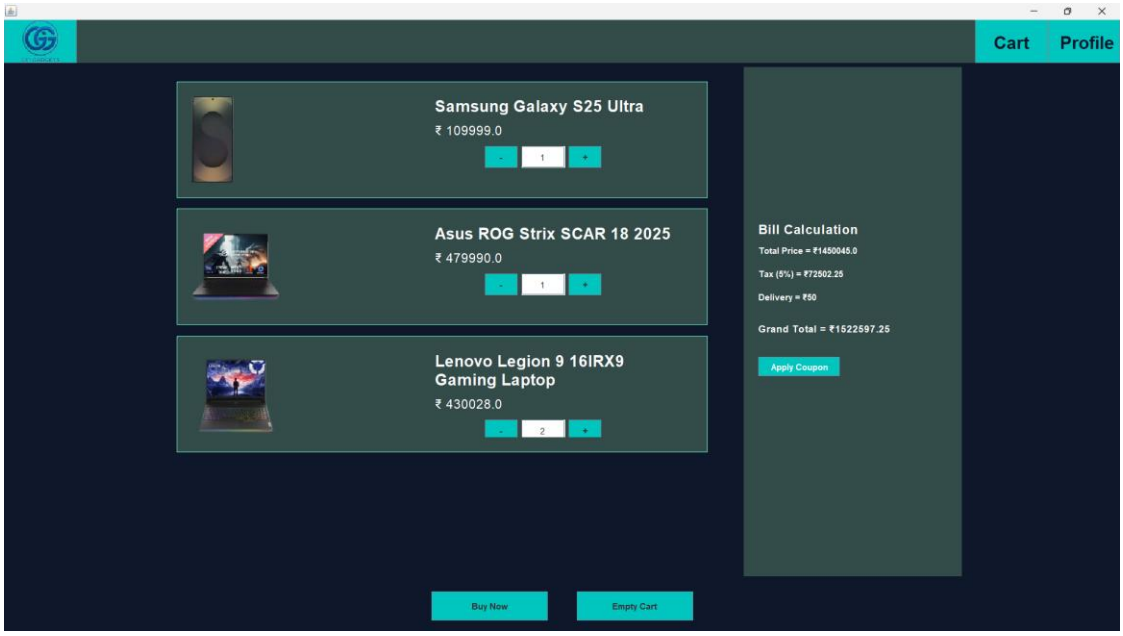

Figure 5: Cart Page

Payment Options: Tthe payment options.

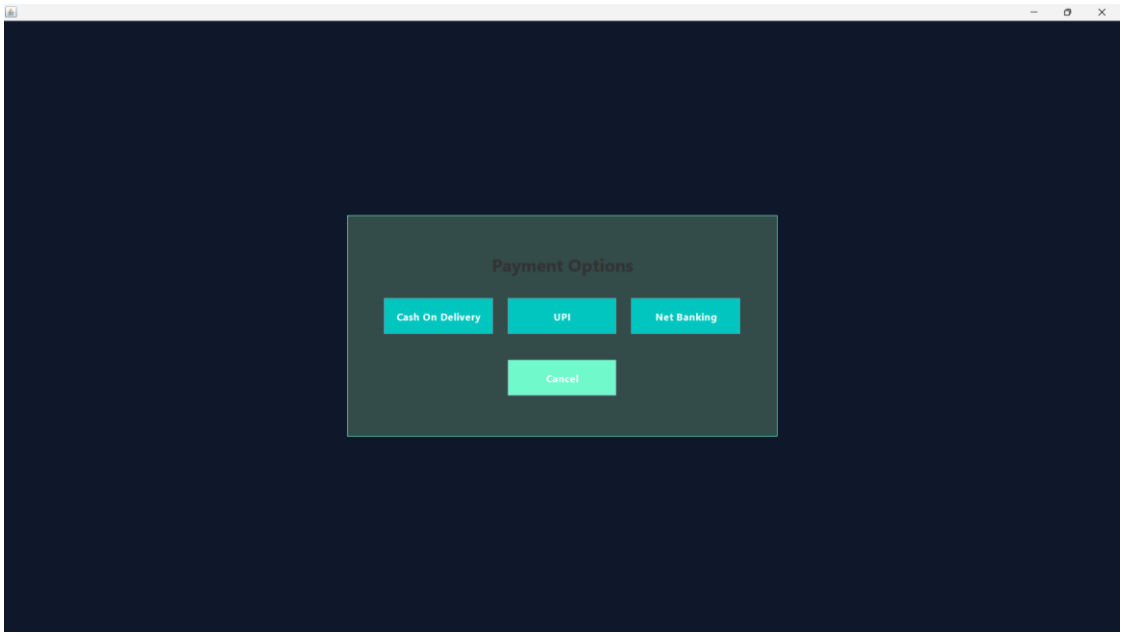

Figure 6: Payment Options

Seller Module:

Add Products: The page that allow the seller add new products.

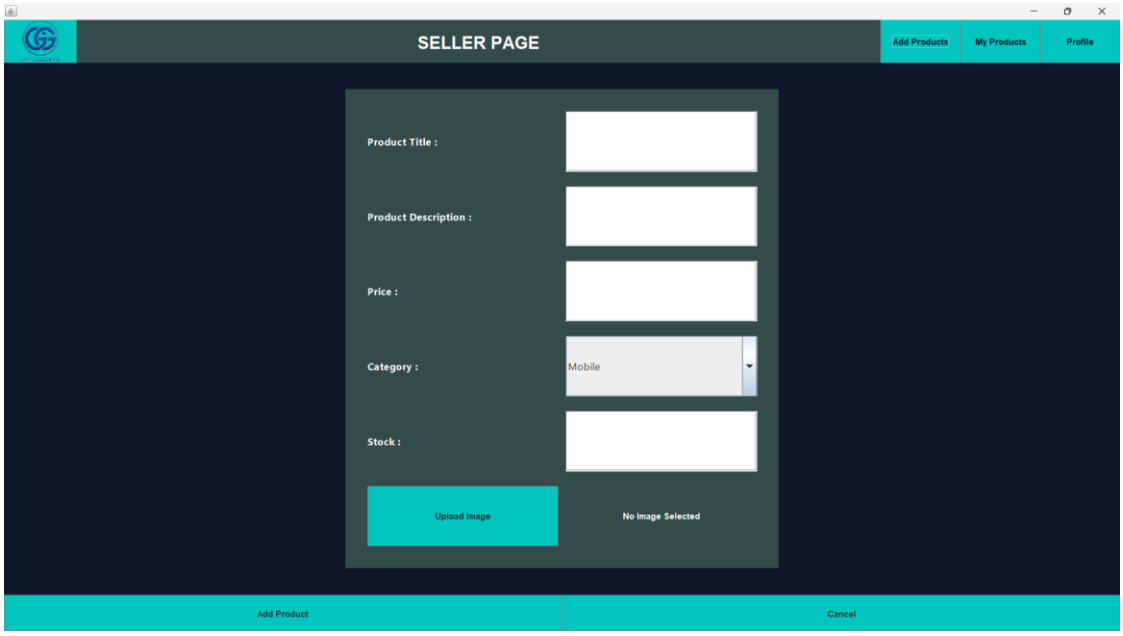

Figure 7: Add product

Edit Products: The page that allow the seller edit already existing products.

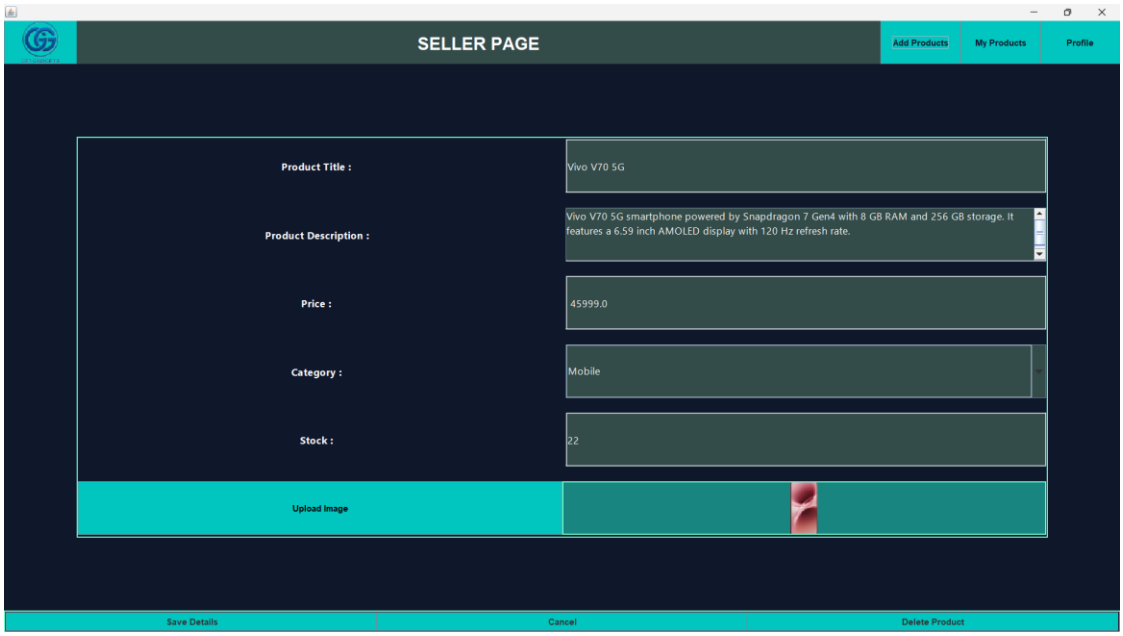

Figure 8: Edit Product

7. CONCLUSION

The E-Commerce Application has been successfully developed to manage product information, cart operations, and order processing in an efficient and organized manner. The system replaces manual or unstructured methods with a computerized solution, thereby reducing errors, saving time, and improving data accuracy.

The application provides essential functionalities such as viewing products, adding items to the cart, managing quantities, placing orders, and applying coupon-based discounts through a user-friendly graphical interface. By using technologies such as Java, Swing, JDBC, and MySQL, the system ensures reliable performance, secure data handling, and smooth interaction between the user interface and the database.

Overall, the system enhances the efficiency of managing product and order data while providing a simplified shopping experience for users. It reduces manual effort and ensures consistency in data storage and retrieval. The modular design of the system also allows it to be easily extended with additional features in the future.

In conclusion, the E-Commerce Application meets the project objectives and serves as an effective solution for demonstrating the core functionalities of an online shopping system.

8. FUTURE ENHANCEMENTS

The E-Commerce Application developed in this project provides basic functionalities for product management, cart operations, and order processing. However, the system can be further enhanced by adding advanced features to improve its functionality and usability.

Some possible future enhancements include:

• **Online Payment Integration:**

Adding secure payment options such as credit/debit cards, UPI, or digital wallets to enable complete online transactions

• **Web-Based Application:**

Converting the desktop application into a web-based system to allow access from anywhere

• **Order Tracking System:**

Providing real-time tracking of orders and delivery status for users

• **Product Recommendation System:**

Suggesting products based on user preferences and purchase history

• **Admin Dashboard:**

Adding a centralized admin panel to manage users, products, and orders efficiently

• **Improved User Interface:**

Enhancing the design and layout for a more modern and responsive user experience

These enhancements can make the system more scalable, secure, and closer to real-world e-commerce platforms.

9. REFERENCES

The following references were used during the development of the Employee Management System:

1. **Herbert Schildt**, *Java: The Complete Reference*, Oracle Press, 11th Edition.
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6. **Java Swing Tutorial – TutorialsPoint**, https://www.tutorialspoint.com/java_swing
7. **W3Schools Java Tutorial**, https://www.w3schools.com/java/

APPENDIX

A.1 Sample Code

Database Connection Code:

The following code establishes a connection between the Java application and the Mysql database using JDBC.

DBConnection.java:

```
Connection getConnection() {
    Connection con = null;
    try {
        Class.forName("com.mysql.cj.jdbc.Driver");
        con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/jdbcdb",
            "root",
            "password"
        );
    } catch (Exception e) {
        e.printStackTrace();
    }
    return con;
}
```

Insertion into cart:

This code is used to add a new products into cart table in the database.

```
    void insertion_into_cart(Cart details){
      String s = "Insert into Cart (user_id, product_id, quantity) values (?, ?, ?)";

      try(Connection con = getConnection();
          PreparedStatement sql = con.prepareStatement(s);){
          sql.setInt(1, details.user_id);
          sql.setInt(2, details.product_id);
          sql.setInt(3, details.quantity);

          sql.executeUpdate();
      }
```

```
        catch(Exception e){
            e.printStackTrace();
        }
    }
```

A.2 SQL Queries

Create users Table:

```
CREATE TABLE users (
    user_id INT AUTO_INCREMENT PRIMARY KEY,
    user_name VARCHAR(50) UNIQUE NOT NULL,
    fullname VARCHAR(100) NOT NULL,
    password VARCHAR(100) NOT NULL,
    address TEXT,
    contact_no VARCHAR(15),
    user_type ENUM('buyer','seller') NOT NULL
);
```

Create products Table:

```
CREATE TABLE products (
    product_id INT AUTO_INCREMENT PRIMARY KEY,
    product_name VARCHAR(100) NOT NULL,
    description TEXT,
    price DOUBLE NOT NULL,
    category VARCHAR(50),
    seller_id INT NOT NULL,
    pic VARCHAR(255),
    stock INT NOT NULL DEFAULT 0,
    rating DOUBLE DEFAULT 0,

    FOREIGN KEY (seller_id) REFERENCES users(user_id)
        ON DELETE CASCADE
);
```